

Angular position θ - rad
Angular velocity

ER - Electromagnetic Radiation

MW - Mechanical Waves (sound)
MT - Material Transfer
ET - Electrical Transmission
Q - Energy (Heat)

FSK CHEAT SHEET

10^{-24}	yocto	y	10^1	deka	da
10^{-21}	zepto	z	10^2	hecto	h
10^{-18}	atto	a	10^3	kilo	k
10^{-15}	femto	f	10^6	mega	M
10^{-12}	pico	p	10^9	giga	G
10^{-9}	nano	n	10^{12}	tera	T
10^{-6}	micro	μ	10^{15}	peta	P
10^{-3}	milli	m	10^{18}	exa	E
10^{-2}	centi	c	10^{21}	zetta	Z
10^{-1}	deci	d	10^{24}	yotta	Y

Motion in 1D:

$$v_{x,avg} = \frac{\Delta x}{\Delta t} = \frac{x_f - x_i}{t_f - t_i}$$

$$v_{avg} = \frac{d}{\Delta t}$$

$$v_{x,inst} = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$$

$$a_{x,inst} = \lim_{\Delta t \rightarrow 0} \frac{\Delta v_x}{\Delta t}$$

$$a_{x,avg} = \frac{\Delta v_x}{\Delta t}$$

$$v_{x,f}^2 = v_{x,i}^2 + 2a\Delta x$$

Vectors:

$$x = r \cos(\theta)$$

$$y = r \sin(\theta)$$

$$\theta = \tan^{-1}\left(\frac{y}{x}\right)$$

$$r = \sqrt{x^2 + y^2}$$

$$\vec{a} \cdot \vec{b} = ab \cos(\theta)$$

Motion in 2D:

$$\vec{v}_{avg} = \frac{\Delta \vec{r}}{\Delta t} = \frac{\Delta x}{\Delta t} \hat{i} + \frac{\Delta y}{\Delta t} \hat{j} + \frac{\Delta z}{\Delta t} \hat{k}$$

$$\vec{a}_{avg} = \frac{\Delta \vec{v}}{\Delta t}$$

$$\vec{a}_{inst} = \frac{dv_x}{dt} \hat{i} + \frac{dv_y}{dt} \hat{j} + \frac{dv_z}{dt} \hat{k}$$

$$\vec{v}_f = \vec{v}_i + \vec{a}t$$

$$\Delta x = v_{ix}t + \frac{1}{2}a_x t^2$$

$$t_{max h} = \frac{v_i \sin \theta_i}{g}$$

$$t_{total} = \frac{2v_i \sin \theta_i}{g}$$

$$h = \frac{v_i^2 \sin^2 \theta_i}{2g}$$

$$R.R. = \frac{v_i^2 \sin 2\theta_i}{g}$$

$$T = \frac{2\pi r}{v}$$

$$\omega = \frac{2\pi}{T}$$

$$v = r\omega$$

$$a_c = \frac{v^2}{r} = (r\omega)^2 = r\omega^2$$

Laws of motion:

$$F_{net} = ma$$

$$\vec{F}_g = G \frac{mM}{r^2} = m\vec{a}$$

$$g = G \frac{M}{r^2}$$

$$G = 6.674 \times 10^{-11}$$

$$\vec{F}_{12} = -\vec{F}_{21}$$

$$F_s = \mu_s F_N$$

$$F_k = \mu_k F_N$$

Circular motion:

$$a_c = \frac{v^2}{r}$$

$$F_{net} = m \frac{v^2}{r}$$

$$T \cos(\theta) = mg$$

$$T \sin(\theta) = \frac{mv^2}{r}$$

$$\tan(\theta) = \frac{v^2}{rg}$$

$$v = \sqrt{rg \tan(\theta)}$$

$$v = \sqrt{Lg \sin(\theta) \tan(\theta)}$$

$$v_{max} = \sqrt{\frac{T_{max} r}{m}}$$

$$F_{s,max} = \frac{mv^2}{r}$$

$$v_{max} = \sqrt{\mu_s g r}$$

$$\theta = \tan^{-1} \frac{v^2}{rg}$$

$$\vec{R} = -b\vec{v}$$

$$R = \frac{1}{2} D r A v^2$$

$$a = g - \left(\frac{D \rho A}{2m} \right) v^2$$

$$V_T = \sqrt{\frac{2mg}{D \rho A}}$$

Energy of a system:

$$W = F \Delta x \cos(\theta)$$

$$W = \int_{x_i}^{x_f} F_x dx$$

$$W_s = \int_{x_i}^{x_f} -kx dx = \frac{1}{2} kx_f^2 - \frac{1}{2} kx_i^2$$

$$\vec{F}_{app} = -\frac{1}{2} kx_{max}^2$$

$$W_{app} = \int_{x_i}^{x_f} kx dx = \frac{1}{2} kx_f^2 - \frac{1}{2} kx_i^2$$

$$F_s = -kx$$

$$K = \frac{1}{2} mv^2$$

$$W_{ext} = \Delta K$$

$$U = mgh$$

$$W_{ext} = \Delta U$$

$$U_s = \frac{1}{2} kx^2$$

$$E_{mech} = K + U$$

Conservation of energy:

$$\Delta E_{system} = \Delta K + \Delta U + \Delta E_{int}$$

$$E_{mech} = \Delta K + \Delta U_g = 0$$

$$E_{mech,f} = E_{mech,i}$$

$$-f_k d = \Delta K$$

$$P_{avg} = \frac{W}{\Delta t}$$

$$P_{inst} = \lim_{\Delta t \rightarrow 0} \frac{W}{\Delta t}$$

$$P = \vec{F} \cdot \vec{v}$$

Linear momentum and collisions:

$$\vec{p} = m\vec{v}$$

$$\vec{p}_{total,f} = \vec{p}_{total,i}$$

$$\vec{I} = \Delta \vec{p} = F_{net} \Delta t$$

Collisions in 1D

$$\text{Elastic coll: } \vec{p}_{total,f} = \vec{p}_{total,i}$$

$$K_i = K_f$$

$$\text{Inelastic coll: } \vec{p}_{total,f} = \vec{p}_{total,i}$$

$$K_i \neq K_f$$

$$\vec{v}_{1,f} = \left[\frac{(m_1 - m_2)}{(m_1 + m_2)} \right] \vec{v}_{1,i} + \left[\frac{(2m_2)}{(m_1 + m_2)} \right] \vec{v}_{2,i}$$

$$\vec{v}_{2,f} = \left[\frac{(2m_1)}{(m_1 + m_2)} \right] \vec{v}_{1,i} + \left[\frac{(m_2 - m_1)}{(m_1 + m_2)} \right] \vec{v}_{2,i}$$

Collisions in 2D

$$m_1 \vec{v}_{1,ix} + m_2 \vec{v}_{2,ix} = m_1 \vec{v}_{1,fx} + m_2 \vec{v}_{2,fx}$$

$$m_1 \vec{v}_{1,iy} + m_2 \vec{v}_{2,iy} = m_1 \vec{v}_{1,fy} + m_2 \vec{v}_{2,fy}$$

Centre of mass

$$x_{CM} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$

$$\vec{v}_{CM} = \frac{1}{M} \sum m_i \vec{v}_i$$

$$\vec{a}_{CM} = \frac{1}{M} \sum m_i \vec{a}_i$$

$$\vec{p}_{ext} = M \vec{v}_{CM}$$

$$\sum \vec{F}_{ext} = M \vec{a}_{CM}$$

$$\vec{I} = \Delta \vec{p}_{total}$$

$$\Delta K + \Delta U + \Delta E_{int} = 0$$

$$W + Q + T_{mech} + T_{int} + T_{ext} + T_{ER}$$

Conical pendulum

$$v_{max} = \sqrt{\frac{N_s n r}{m}}$$

Resistive force

Rotation:

(Equations of motion are the same)

$$x = \theta = \frac{s}{r}$$

$$v_{avg} = \omega_{avg} = \frac{\Delta \theta}{\Delta t}$$

$$\omega_{inst} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \theta}{\Delta t}$$

$$a_{avg} = \alpha_{avg} = \frac{\Delta \omega}{\Delta t}$$

$$\alpha_{inst} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \omega}{\Delta t}$$

$$T = \frac{2\pi}{\omega} = \frac{1}{f}$$

$$f = \frac{\omega}{2\pi} = \frac{1}{T}$$

$$v = \omega r$$

$$a_r = \frac{v^2}{r} = \omega^2 r^2$$

o **Torque:**

$$\tau = Fr \sin \phi$$

$$I = mr^2$$

Hoop/thin cylindrical shell	$I_{CM} = MR^2$
Hollow cylinder	$I_{CM} = \frac{1}{2} M(R_1^2 + R_2^2)$
Solid cylinder/disk	$I_{CM} = \frac{1}{2} MR^2$
Rectangular plate	$I_{CM} = \frac{1}{12} M(a^2 + b^2)$
Long thin rod with rotation axis through center	$I_{CM} = \frac{1}{12} M(a^2 + b^2)$
Long thin rod with rotation axis through end	$I_{CM} = \frac{1}{3} ML^2$
Solid sphere	$I_{CM} = \frac{2}{5} MR^2$
Thin spherical shell	$I_{CM} = \frac{2}{3} MR^2$

$$I = I_{CM} + MD^2$$

$$K = \frac{1}{2} (\sum mr^2) \omega^2$$

$$I = \sum mr^2$$

$$K = \frac{1}{2} I \omega^2$$

Oscillatory motion:

$$F = -kx$$

$$a_x = -\frac{k}{m}x$$

$$\omega = \frac{2\pi}{T} = 2\pi f$$

$$x(t) = A \cos(\omega t + \phi) \rightarrow m$$

$$v(t) = -\omega A \sin(\omega t + \phi) \rightarrow m/s$$

$$a(t) = -\omega^2 x(t) \rightarrow m/s^2$$

$$\omega t + \phi \rightarrow \text{rad}$$

$$f = \frac{\omega}{2\pi}$$

$$T = \frac{1}{f} = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{m}{k}}$$

$$U(t) = \frac{1}{2} kA^2 \cos^2(\omega t + \phi)$$

$$K(t) = \frac{1}{2} kA^2 \sin^2(\omega t + \phi)$$

$$E_{mech} = U + K = \frac{1}{2} kA^2$$

$$v = \sqrt{\frac{k}{m}(A^2 - x^2)} = \pm \omega \sqrt{A^2 - x^2}$$

$$\omega = \sqrt{\frac{g}{L}}$$

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{L}{g}}$$

$$\tau = -mgd \sin \theta$$

$$\omega = \sqrt{\frac{mgd}{I}}$$

$$T = 2\pi \sqrt{\frac{I}{mgd}}$$

$$T = 2\pi \sqrt{\frac{1}{mgd}}$$

Wave motion:

$$y(x, t) = A \sin(kx - \omega t + \phi)$$

$$y(x, t) = A \sin\left[2\pi\left(\frac{x}{\lambda} - \frac{t}{T}\right)\right]$$

$$k = \frac{2\pi}{\lambda}$$

$$\omega = \frac{2\pi}{T} = 2\pi f$$

$$v = \lambda f = \frac{\omega}{k}$$

$$v_y = -\omega A \cos(kx - \omega t)$$

$$v_{y,max} = \omega A$$

$$a_y = -\omega^2 A \sin(kx - \omega t)$$

$$a_{y,max} = \omega^2 A$$

$$v = \sqrt{\frac{T}{\mu}}$$

Superposition:

$$y'(x, t) = y_1(x, t) + y_2(x, t)$$

$$y'(x, t) = \left[2A \cos \frac{\phi}{2}\right] \sin kx - \omega t + \frac{\phi}{2}$$

Standing waves:

$$y'(x, t) = (2A \sin kx) \cos \omega t$$

$$x = n \frac{\lambda}{2} \rightarrow \text{nodes}$$

$$kx = \left(n + \frac{1}{2}\right)\pi \rightarrow \text{anti-nodes}$$

$$\lambda = \frac{2L}{n}$$

$$f = \frac{v}{\lambda} = n \frac{v}{2L}$$

Temperature:

$$P = P_0 + \rho gh$$

$$T_C = T_K - 273.15$$

$$\Delta L = L_0 \alpha \Delta T$$

$$\Delta V = V \beta \Delta T$$

$$\beta = 3\alpha$$

$$PV = nRT = \frac{N}{N_A} RT$$

$$PV = Nk_B T$$

$$k_B = \frac{R}{N_A} = 1.38 \times 10^{-23}$$

Thermodynamics:

$$Q = C \Delta T$$

$$Q = cm \Delta T$$

$$Q_{cold} = -Q_{hot}$$

$$m_w c_w (T_f - T_w) = -m_s c_s (T_f - T_s)$$

$$Q = L \Delta m$$

$$\Delta E_{int} = W + Q$$

$$F = PA$$

$$W = - \int_{V_i}^{V_f} P dV$$

o **Constant-pressure followed by constant-volume:**

$$W = -P_i (V_f - V_i)$$

o **Constant-volume followed by constant-pressure:**

$$W = -P_f (V_f - V_i)$$

o **An arbitrary compression:**

$$W = - \int_{V_i}^{V_f} P dV$$

$$\Delta E_{int} = W + Q = 0 \rightarrow \text{Isolated system}$$

$$W = nRT \ln \left(\frac{V_f}{V_i}\right) \rightarrow \text{Isothermal process}$$

$$\Delta E_{int} = W \rightarrow \text{Adiabatic process}$$

$$\Delta E_{int} = 0$$

$$\Delta E_{int} = Q \rightarrow \text{Isovolumetric process}$$

$$W = -P(V_f - V_i) \rightarrow \text{Isobaric process}$$

$$P = kA \left|\frac{dT}{dx}\right|$$

$$\left|\frac{dT}{dx}\right| = \frac{T_h - T_c}{L}$$

$$P = kA \left(\frac{T_h - T_c}{L}\right)$$

$$P = \frac{A(T_h - T_c)}{\sum_i \left(\frac{L_i}{k_i}\right)} = \frac{A(T_h - T_c)}{\sum_i R_i}$$

o **Radiation**

$$P = \sigma A e T^4$$

$$P_{net} = \sigma A e (T^4 - T_0^4)$$